

Effect of cocoa/chocolate ingestion on brachial artery flow-mediated dilation and its relevance to cardiovascular health and disease in humans.

Prospective studies indicate that high intake of dietary flavanols, such as those contained in cocoa/chocolate, are associated with reduced rates of cardiovascular-related morbidity and mortality in humans. Numerous mechanisms may underlie these associations such as favorable effects of flavanols on blood pressure, platelet aggregation, thrombosis, inflammation, and the vascular endothelium. The brachial artery flow-mediated dilation (FMD) technique has emerged as a robust method to quantify endothelial function in humans. Collectively, the preponderance of evidence indicates that FMD is a powerful surrogate measure for firm cardiovascular endpoints, such as cardiovascular-related mortality, in humans. Thus, literally thousands of studies have utilized this technique to document group differences in FMD, as well as to assess the effects of various interventions on FMD. In regards to the latter, numerous studies indicate that both acute and chronic ingestion of cocoa/chocolate increases FMD in humans. Increases in FMD after cocoa/chocolate ingestion appear to be dose-dependent such that greater increases in FMD are observed after ingestion of larger quantities. The mechanisms underlying these responses are likely diverse, however most data suggest an effect of increased nitric oxide bioavailability. Thus, positive vascular effects of cocoa/chocolate on the endothelium may underlie (i.e., be linked mechanistically to) reductions in cardiovascular risk in humans. Copyright © 2012 Elsevier Inc. All rights reserved.